

compute gradients for (activations + parameters)

Backward Flops: $\sim 2-4 \times$ more expensive than forward Flops

$x_1 \rightarrow$ input (B, D) $\xrightarrow{\text{Batch}}$ shape (input/hidden dim) $\rightarrow$ shape $\downarrow$

$w_1 \rightarrow$ first layer weights (D, D)

$h_1 = x @ w_1 \rightarrow$ hidden activation after first layer (B, D) $\xrightarrow{\text{Batch}}$ dim $\rightarrow$ dim

$w_2 \rightarrow$ second layer weights (D, K) $\rightarrow$ shape

$h_2 = h_1 @ w_2 \rightarrow$ pre-loss activation (B, K) $\rightarrow$ shape

$\text{loss} = h_2 \cdot \text{pow}(2). \text{mean}()$ $\rightarrow$ scalar $\xrightarrow{\text{Batch}}$

$\Rightarrow h_1/h_2$ are intermediate activations, not learnable params

e.g. $h_1 = x @ w_1$
 $h_2 = h_1 @ w_2$
 $\text{loss} = f(h_2)$

$$\Rightarrow \frac{d\text{loss}}{dw_1} = \frac{d\text{loss}}{dh_2} * \frac{dh_2}{dh_1} * \frac{dh_1}{dw_1} \quad \text{Chain rules}$$

$$\begin{cases} h_1.\text{grad} = \frac{d\text{loss}}{dh_1} \\ h_2.\text{grad} = \frac{d\text{loss}}{dh_2} \\ w_1.\text{grad} = \frac{d\text{loss}}{dw_1} \\ w_2.\text{grad} = \frac{d\text{loss}}{dw_2} \end{cases} \Rightarrow \begin{aligned} h_1.\text{grad}[i, j] &= \sum_k w_2[i, k] * h_2.\text{grad}[i, k] \\ w_2.\text{grad}[j, k] &= \sum_{i'} h_1[i', j] * h_2.\text{grad}[i', k] \end{aligned}$$

$\downarrow$ why? $\downarrow$ output dim $\downarrow$ Batch $\downarrow$ sample index

(chain rule)

$$\Rightarrow \frac{d\text{loss}}{dw_2[n, k]} = \sum_i \left(\frac{d\text{loss}}{dh_2[i, k]} \right) * \frac{dh_2[i, k]}{dw_2[n, k]}$$

$$\begin{aligned} h_2[i, k] &= \sum_n h_1[i, n] * w_2[n, k] \\ \frac{dh_2[i, k]}{dw_2[n, k]} &= h_1[i, n] \end{aligned}$$

for a linear model: currency flops

$$X \xrightarrow{W_1} h_1 \xrightarrow{W_2} h_2 \rightarrow \text{loss}$$

$h_1 = X @ W_1 \Rightarrow \frac{dh_1}{dW_1} = h_1$ hidden #2
 $h_2 = h_1 @ W_2 \Rightarrow \frac{dh_2}{dh_1} = W_2$ hidden #1 weight
 $h_2[i, k] = \sum_n h_1[i, n] \cdot W_2[n, k]$
 $\text{loss} = h_2 \cdot \text{pow}(2)_{\text{mean}(i)}$

(#1) $\downarrow$
 for W_2 grad, sum over batch index (i): how much should weight $j \rightarrow k$ change? look across all batch samples i .
 for hidden grad, sum over outputs (k): how much did hidden unit j contribute to errors? look across outputs k .

for (#1): $W_2 \cdot \text{grad}[i, k] = \sum_j h_1[i, j] * h_2 \cdot \text{grad}[j, k]$
 $\downarrow$
 update weight from hidden $j \rightarrow$ output k , summing contributions from all batch samples

for (#2): $h_1 \cdot \text{grad}[i, j] = \sum_k W_2[i, k] * h_2 \cdot \text{grad}[i, k]$
 $\downarrow$
 How much hidden unit j contributed to the error in all outputs for sample i .

for equation (#1): $W_2 \cdot \text{grad}$ = ^{so scoring} update the rule based on all student's results

To deduce the equation (#1):

what we have $\Rightarrow \frac{\partial h_2[i, k]}{\partial W_2[n, k]} = h_1[i, n]$ using
 $\frac{\partial L}{\partial h_2[i, k]} = h_2 \cdot \text{grad}[i, k]$